Agile Model

Agile model

- Agile Software Development is an iterative and incremental approach to Software Development that emphasizes the importance of delivering a working product quickly and frequently.
- It involves close collaboration between the development team and the customer to ensure that the product meets their needs and expectations.

Why is Agile Used?

- Agile is used because it helps teams deliver value quickly and continuously.
- By prioritizing the delivery of difficult results early in the project, customers benefit from seeing and using the product sooner, allowing for quick feedback and adjustments.
- Agile also encourages teams to focus on what truly matters, concentrating on tasks that add value and avoiding unnecessary work.

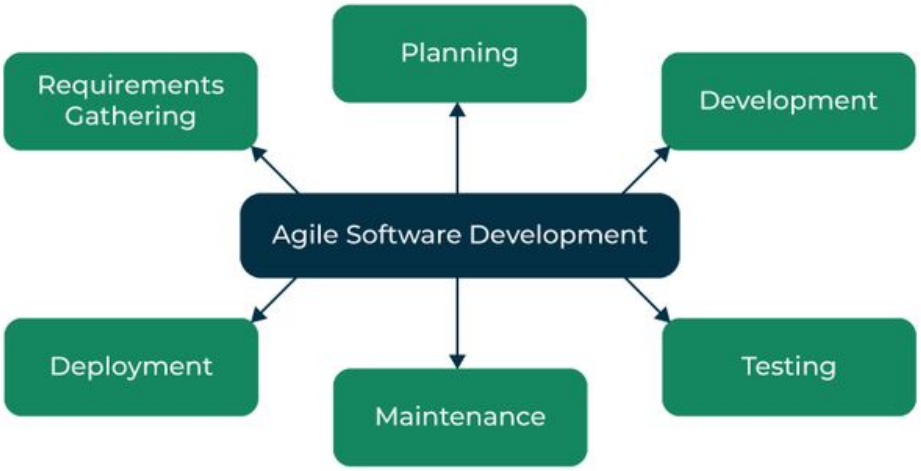
Why is Agile Used?

- **Agile as a Mindset:** Agile represents a shift in culture that values adaptability, collaboration, and client happiness. It gives team members more authority and promotes a cooperative and upbeat work atmosphere.
- **Quick Response to Change:** Agile fosters a culture that allows teams to respond swiftly to constantly shifting priorities and requirements. This adaptability is particularly useful in sectors of the economy or technology that experience fast changes.
- **Regular Demonstrations:** Agile techniques place a strong emphasis on regular demonstrations of project progress. Stakeholders may clearly see the project's status, upcoming problems, and upcoming new features due to this transparency.
- **Cross-Functional Teams:** Agile fosters self-organizing, cross-functional teams that share information effectively, communicate more effectively and feel more like a unit.

Agile Software Development Process

- Agile software development, often just called Agile, focuses on being flexible and practical when delivering software.
- Instead of launching everything at once, Agile delivers small, valuable updates to users over time.
- This approach allows teams to adjust and improve the product along the way, verifying that each update brings real value to the users.
- It’s all about making progress in manageable steps and responding quickly to changes.

Agile Software Development Process



Agile Software Development Process

- **1. Requirements Gathering**
- This is the first step where the development team works closely with the customer to understand what they really need from the software. The team listens carefully to the customer’s needs, then sorts and prioritizes these requirements to make sure the most important features are developed first.
- **2. Planning**
- In this stage, the team creates a clear plan for how they’ll build the software. They decide which features to focus on in each development cycle (called an iteration). Think of it like mapping out the journey of the project, so everyone knows what to expect and when things will be delivered.

Agile Software Development Process

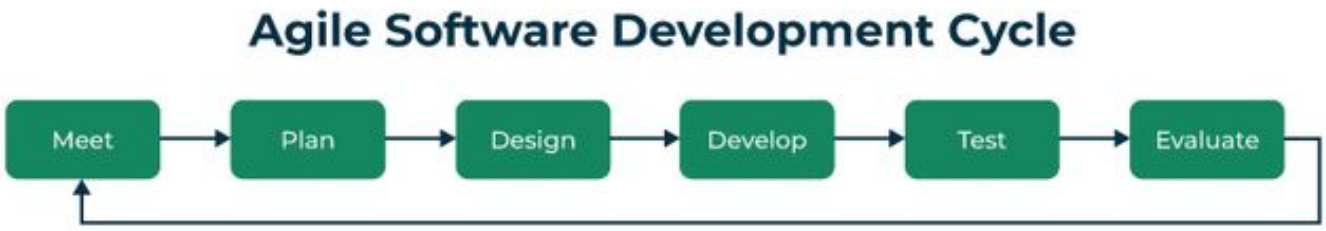
- **3. Development**
- This is where the team starts turning their plan into reality. They work in short, focused cycles, building small, usable pieces of the product. Each cycle builds on the last, which helps the team stay on track and get quick feedback to keep improving.
- **4. Testing**
- As the software gets built, it’s also tested to make sure it works properly and meets the customer’s needs. Testing ensures the product is of high quality and free from errors, so problems are caught early on before they become bigger issues.

Agile Software Development Process

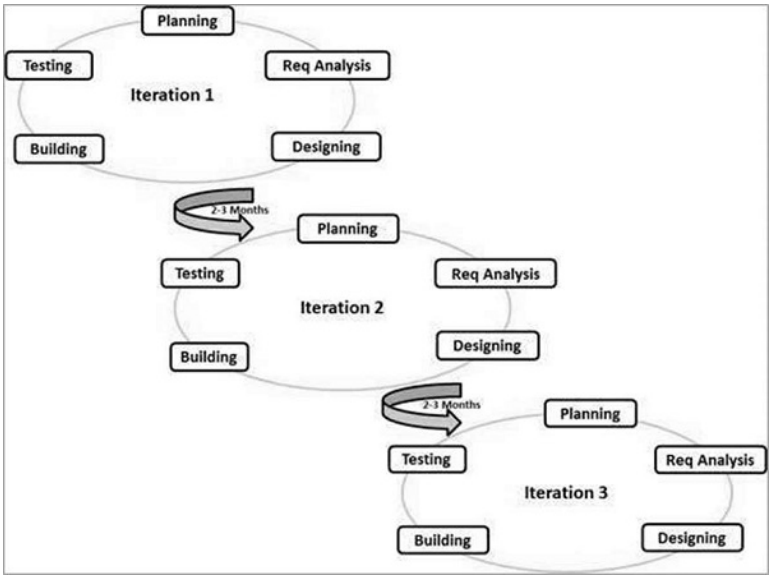
- **5. Deployment**
- Once everything is tested and working as expected, the software is deployed, which means it’s ready for customers or end-users to start using. It’s the moment when all the development work comes to life.
- **6. Maintenance**
- Even after the software is released, the job isn’t done. The team keeps maintaining the software, verifying it continues to work well and stays up-to-date with any new needs or changes from the customer. This keeps the software relevant and helpful over time.

- Agile Software Development is widely used by software development teams and is considered to be a flexible and adaptable approach to software development that is well-suited to changing requirements and the fast pace of software development.
- Agile is a time-bound, iterative approach to software delivery that builds software incrementally from the start of the project, instead of trying to deliver all at once.

Agile Software Development Cycle



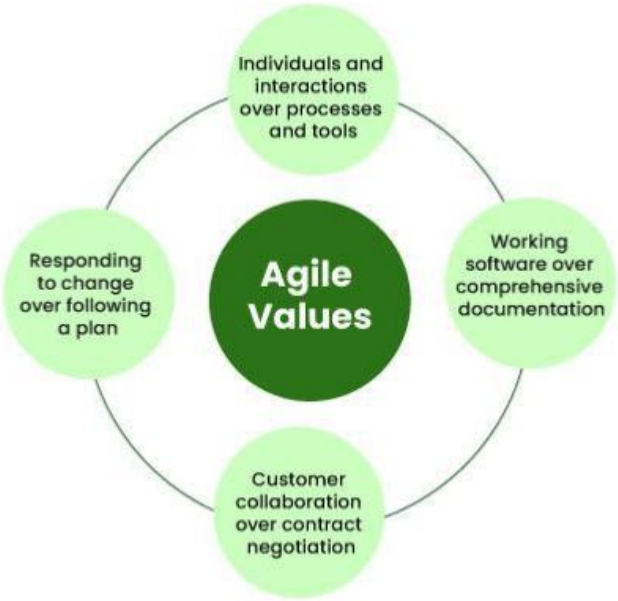
Agile Software Development Cycle



Agile Software Development Cycle

- Step 1: In the first step, concept, and business opportunities in each possible project are identified and the amount of time and work needed to complete the project is estimated. Based on their technical and financial viability, projects can then be prioritized and determined which ones are worthwhile pursuing.
- Step 2: In the second phase, known as inception, the customer is consulted regarding the initial requirements, team members are selected, and funding is secured. Additionally, a schedule outlining each team's responsibilities and the precise time at which each sprint's work is expected to be finished should be developed.
- Step 3: Teams begin building functional software in the third step, iteration/construction, based on requirements and ongoing feedback. Iterations, also known as single development cycles, are the foundation of the Agile software development cycle.

4 Core Values of Agile Software Development



4 Core Values of Agile Software Development

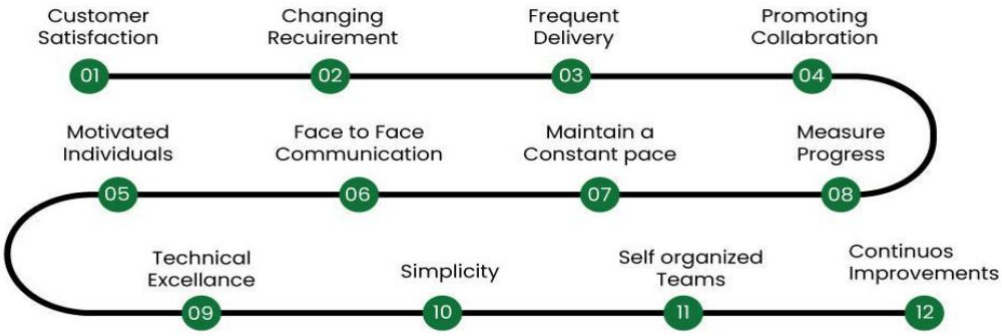
- 1. **Individuals and Interactions over Processes and Tools:** This value stresses that the strength of the team and how well they work together is more important than the tools or processes they use. Of course, tools and processes help, but the real success of a project comes from good teamwork, open communication, and collaboration.
- 2. **Working Software over Comprehensive Documentation:** Agile prefers delivering functional software quickly rather than getting down in lengthy documentation. While some documentation is needed, the focus is on getting a working product into the hands of the user and improving it based on feedback.

4 Core Values of Agile Software Development

- **3. Customer Collaboration over Contract Negotiation:** Agile values regular collaboration with customers over sticking strictly to contracts. The idea is to involve the customer throughout the development, verifying that the product meets their needs and making adjustments based on their feedback.
- **4. Responding to Change over Following a Plan:** In Agile, change is expected, and the approach encourages flexibility. Rather than rigidly following a plan that may no longer apply, Agile teams adapt and adjust based on new information, changing market conditions, or including customer requirements.

12 Principles of Agile Software Development

- Agile principles are guidelines for flexible and efficient software development. They emphasize frequent delivery, embracing change, collaboration, and continuous improvement. The focus is on delivering value, maintaining a sustainable work pace, and ensuring technical excellence.



These principles include:

- Ensuring customer satisfaction through the early delivery of software.
- Being open to changing requirements in the stages of the development.
- Frequently delivering working software with a main focus on preference for timeframes.
- Promoting collaboration between business stakeholders and developers as an element.
- Structuring the projects around individuals. Providing them with the necessary environment and support.

These principles include:

- Prioritizing face to face communication whenever needed.
- Considering working software as the measure of the progress.
- Fostering development by allowing teams to maintain a pace indefinitely.
- Placing attention on excellence and good design practices.
- Recognizing the simplicity as crucial factor aiming to maximize productivity by minimizing the work.
- Encouraging self organizing teams as the approach to design and build systems.
- Regularly reflecting on how to enhance effectiveness and to make adjustments accordingly.

Advantages Agile Software Development

- Increased collaboration and communication: Agile Software Development Methodology emphasize collaboration and communication among team members, stakeholders, and customers. This leads to improved understanding, better alignment, and increased buy-in from everyone involved.
- Flexibility and adaptability: Agile methodologies are designed to be flexible and adaptable, making it easier to respond to changes in requirements, priorities, or market conditions. This allows teams to quickly adjust their approach and stay focused on delivering value.
- Improved quality and reliability: Agile methodologies place a strong emphasis on testing, quality assurance, and continuous improvement. This helps to ensure that software is delivered with high quality and reliability, reducing the risk of defects or issues that can impact the user experience.

Advantages Agile Software Development

- Enhanced customer satisfaction: Agile methodologies prioritize customer satisfaction and focus on delivering value to the customer. By involving customers throughout the development process, teams can ensure that the software meets their needs and expectations.
- Increased team morale and motivation: Agile methodologies promote a collaborative, supportive, and positive work environment. This can lead to increased team morale, motivation, and engagement, which can in turn lead to better productivity, higher quality work, and improved outcomes.

Advantages Agile Software Development

- Deployment of software is quicker and thus helps in increasing the trust of the customer.
- Can better adapt to rapidly changing requirements and respond faster.
- Helps in getting immediate feedback which can be used to improve the software in the next increment.
- People - Not Process. People and interactions are given a higher priority than processes and tools.
- Continuous attention to technical excellence and good design.

Disadvantages Agile Software Development

- Lack of predictability: Agile Development relies heavily on customer feedback and continuous iteration, which can make it difficult to predict project outcomes, timelines, and budgets.
- Limited scope control: Agile Development is designed to be flexible and adaptable, which means that scope changes can be easily accommodated. However, this can also lead to scope creep and a lack of control over the project scope.
- Lack of emphasis on testing: Agile Development places a greater emphasis on delivering working code quickly, which can lead to a lack of focus on testing and quality assurance. This can result in bugs and other issues that may go undetected until later stages of the project.

Disadvantages Agile Software Development

- Risk of team burnout: Agile Development can be intense and fast-paced, with frequent sprints and deadlines. This can put a lot of pressure on team members and lead to burnout, especially if the team is not given adequate time for rest and recovery.
- Lack of structure and governance: Agile Development is often less formal and structured than other development methodologies, which can lead to a lack of governance and oversight. This can result in inconsistent processes and practices, which can impact project quality and outcomes.
- In the case of large software projects, it is difficult to assess the effort required at the initial stages of the software development life cycle.

Disadvantages Agile Software Development

- Agile Development is more code-focused and produces less documentation.
- Agile development is heavily dependent on the inputs of the customer. If the customer has ambiguity in his vision of the outcome, it is highly likely that the project to get off track.
- Face-to-face communication is harder in large-scale organizations.
- Only senior programmers are capable of making the kind of decisions required during the development process. Hence, it's a difficult situation for new programmers to adapt to the environment.

Practices of Agile Software Development

- Scrum: Scrum is a framework for agile software development that involves iterative cycles called sprints, daily stand-up meetings, and a product backlog that is prioritized by the customer.
- Kanban: Kanban is a visual system that helps teams manage their work and improve their processes. It involves using a board with columns to represent different stages of the development process, and cards or sticky notes to represent work items.
- Continuous Integration: Continuous Integration is the practice of frequently merging code changes into a shared repository, which helps to identify and resolve conflicts early in the development process.

Practices of Agile Software Development

- Test-Driven Development: Test-Driven Development (TDD) is a development practice that involves writing automated tests before writing the code. This helps to ensure that the code meets the requirements and reduces the likelihood of defects.
- Pair Programming: Pair programming involves two developers working together on the same code. This helps to improve code quality, share knowledge, and reduce the likelihood of defects.